Lab II: HDFS Hands-on Exercises

1. Use HDFS commands for the following tasks:
   1) Given a directory in HDFS, output the following information of all files in the directory: read and write permissions, file size and creation time.
   2) Display the content of a file in HDFS.
   3) Copy a text file from local file system to HDFS; if the file already exists in HDFS, append the content to the end of the existing file.
   4) Copy a file from HDFS to local file system.
   5) Delete a file in HDFS.

2. Write a Java program using Hadoop Java API for the following tasks:
   1) Copy a text file from local file system to HDFS; if the file already exists in HDFS, let the user specify whether to append the content to the end of the existing file, or overwrite the original file.
   2) Copy a file from HDFS to local file system; if there is a file with the same name in local file system, then rename it (the new file) automatically

3. Report
   1) State your experimental environment (OS, JDK, Hadoop version etc.).
   2) Show how you conduct the experiment (step by step) and show your codes/commands, results, and screenshots.
   3) List the problems you have in the experiments, and how you find the solutions, if any.

4. References
   1) HDFS commands: http://hadoop.apache.org/docs/stable/hadoop-project-dist/hadoop-common/FileSystemShell.html
   2) Hadoop Java API: http://hadoop.apache.org/docs/stable/api/index.html